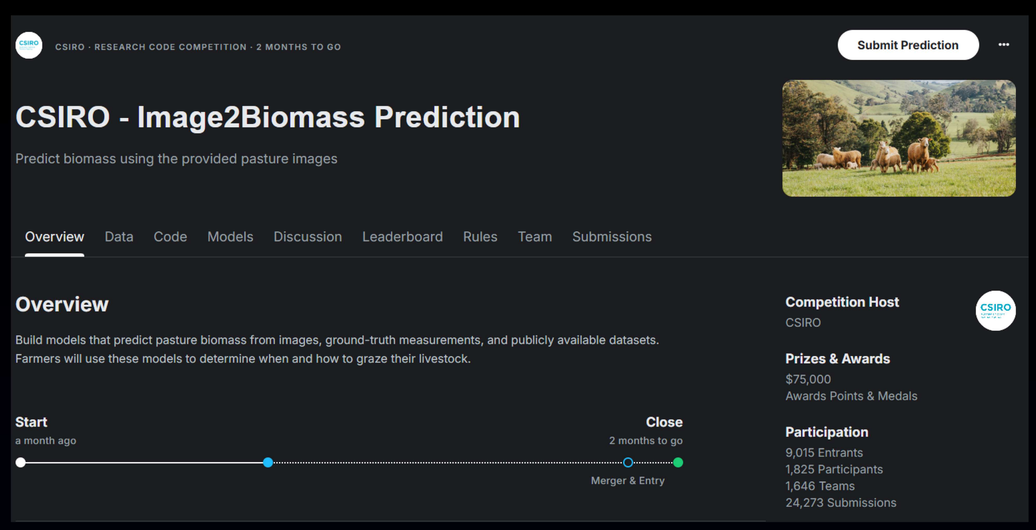

CSIRO IMAGE2BIOMASS PREDICTION

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Introduction

Efficient livestock management relies on accurate pasture biomass estimation. Conventional methods are often slow, labor-intensive, and difficult to scale across varying terrain conditions.

To develop an accurate deep learning model to estimate pasture biomass from images, facilitating better grazing management. This project is based on the CSIRO Kaggle Competition.

RESULT

MODEL	VAL RSME (AVG)	LOSS FUNCTION	BEST SCORE (MAX 1.0)
RESNET50	MSE LOSS	~22.3	0.36
EFFICIENTNET-B4	HUBER LOSS	~12.9	0.40
EFFICIENTNET-B4 + XGB	SQUARED ERROR	~18.9	0.38
DINOV2-GIANT + SIGLIP	L2 SQUARED	~14.93	0.59

We evaluated the models based on Validation RMSE (Root Mean Square Error).

Huber Loss (delta=1.0) and Cosine Annealing scheduler significantly improved stability compared to the baseline MSE.

CONCLUSSION

The End-to-End approach using EfficientNet-B4 with robust loss functions outperforms hybrid or ensemble methods for this specific biomass estimation task. The model provides a fast and scalable alternative to manual pasture measurement.

METHODOLOGY

AUGMENTATION → NORMALIZATION → MODEL

PREPROCESSING & AUGMENTATTION

To ensure model robustness against lighting and terrain variations, we applied:

- Augmentation: Random Rotation (30°), Horizontal/Vertical Flip, Color Jitter.
- Normalization: ImageNet standards.
- Resizing: 224×224 (ResNet) and 380×380 (EfficientNet).

EXPERIMENTAL SCENARIOS

We tested four distinct architectural approaches:

- ResNet50: Standard Transfer Learning (Baseline).
- EfficientNet-B4: High-Res input with Huber Loss (End-to-End).
- EffNet-B4 + XGBoost: Deep Feature Extraction approach.
- DINOv2-Giant + SigLIP: Zero-shot Vision Feature Extraction + Ensemble Regression (LGBM + GBR + Ridge).

DATASET & OBJECTIVES

The dataset consists of pasture images with 5 continuous target classes based on biomass weight:

- Dry_Green_g (0.1)
- Dry_Dead_g (0.1)
- Dry_Clover_g (0.1)
- GDM_g (0.2)
- Dry_Total_g (0.5)